Stack and Queue

→ discuss little on programming and problem solving.

How different DS are helpful in solving problem and do particular task fast.

Choice of DS ⇒ efficient performance

Stack → Linear DS

Sequence and position matters,
↳
might not be physically but logically are
↳ like linked list

Now, example

1. Stack of Plates, books etc.

2. Pringles





Explain these properly







Now, explain what is Last in first out (LIFO)

Abstract Data Structure : ADT is a type for objects whose behaviour is defined by set of operation.
as abstract, no knowledge of how are they performed

We will be implementing using Array list and linked list both.

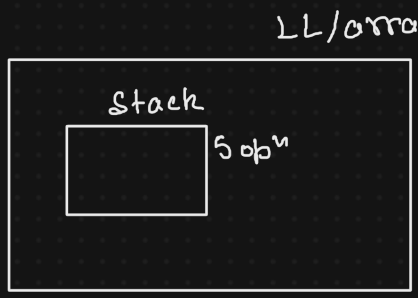
Industry Example

→ Call stack in Programming
→ Undo & Redo in Editors
→ Matching HTML tags in web dev

Operation on Stack

Explain with Pringle example

1. Peek / Top
2. Push
3. Pop
4. Size
5. isempty()



If some extra opⁿ, required then not a stack

If you don't want to use then other DS

Also will make sure that these opⁿ are optimized.

Stack using Inbuilt List



Also mention how we can though still access list ideally won't as we have stack main and/or make private

Also tell that how we are going to follow this and a direct list going forward.

Stack using a fixed size list

Explain about [None] * 5
& also about the top variable and size.

also explain how size is not needed but will be calculated via top

Also for dynamic array (Question and then Answer)

Stack using linked list

First explain using the normal LL implementation and then why O(n) for pop is not good. (with tail)

Then next video discuss on LL from head

Then finally for our implementation